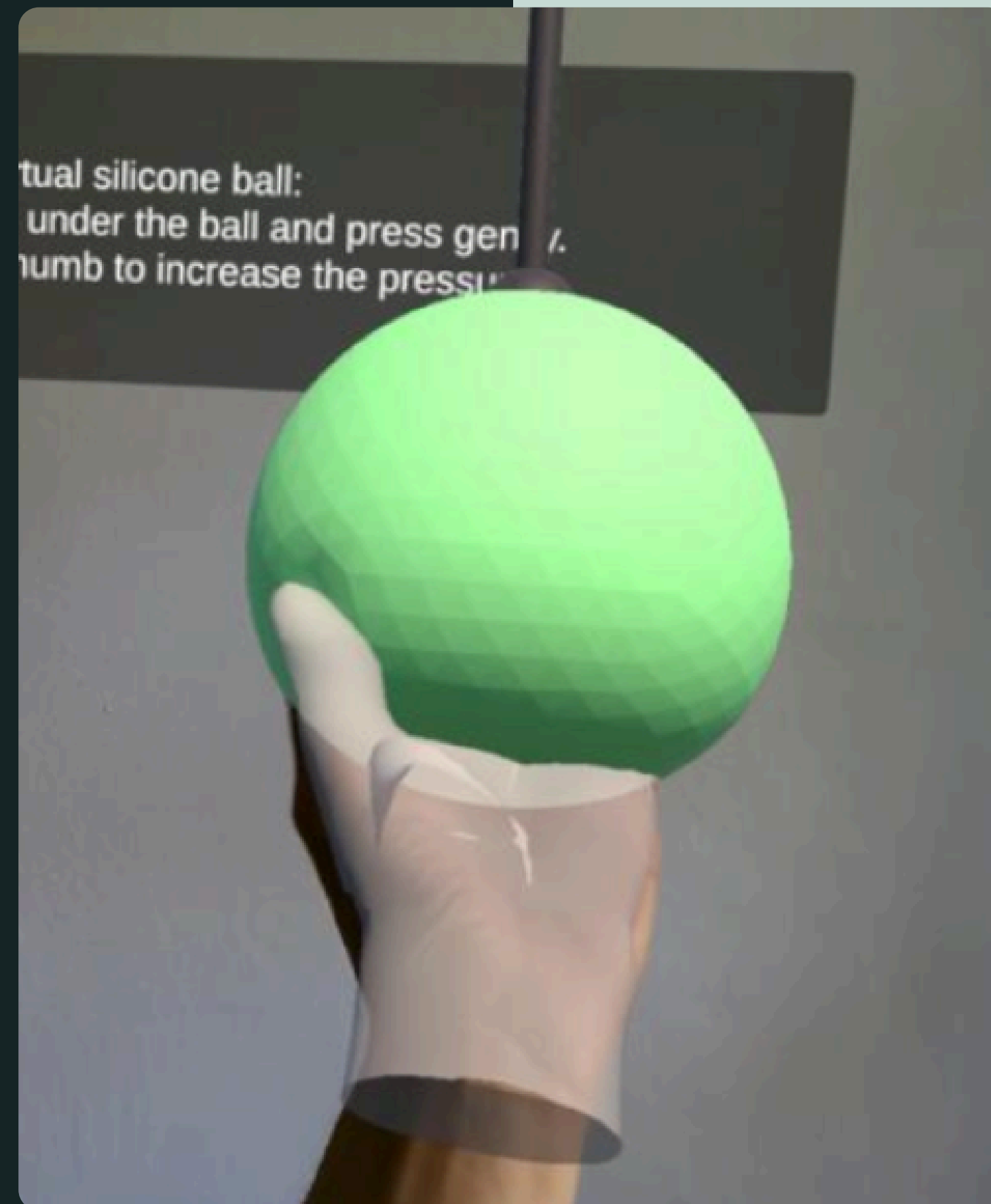


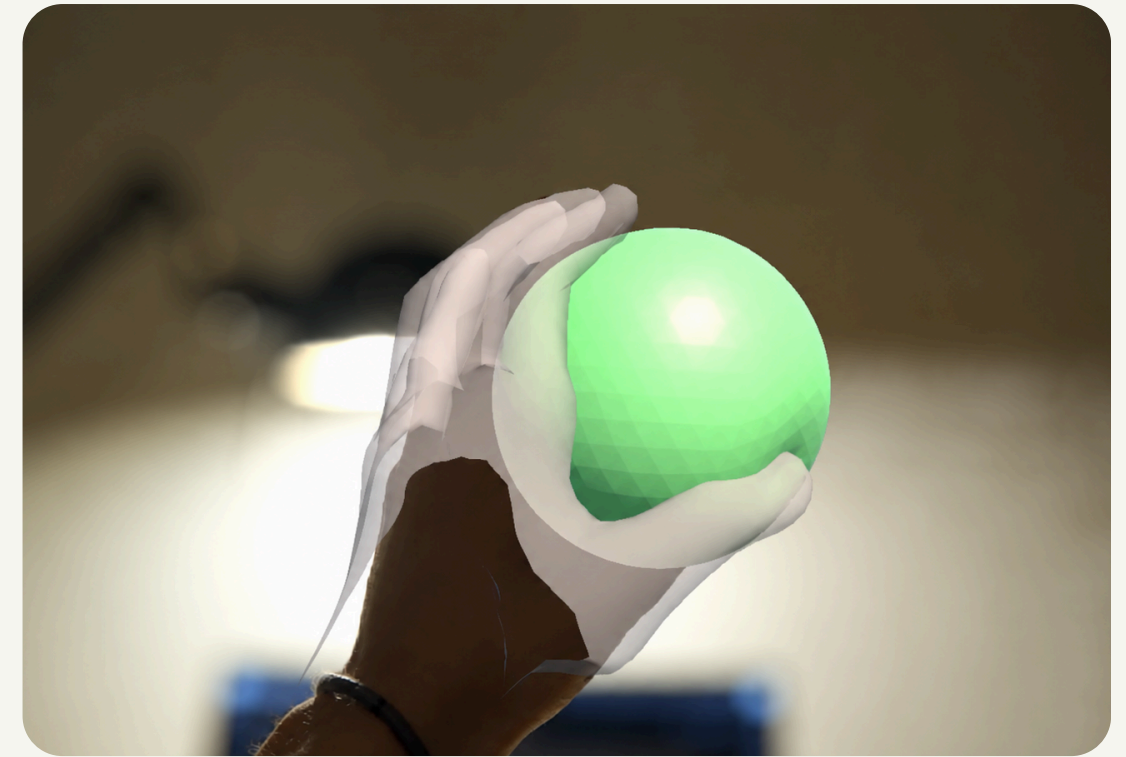
pseudo-haptic

VISUAL DEFORMATION FOR PSEUDO-HAPTIC
SOFTNESS PERCEPTION IN AR.



Research Problem

This pilot study investigates whether visual manipulation alone can induce a pseudo-haptic perception of softness during interaction with a purely virtual object, and therefore with an object that is physically intangible.

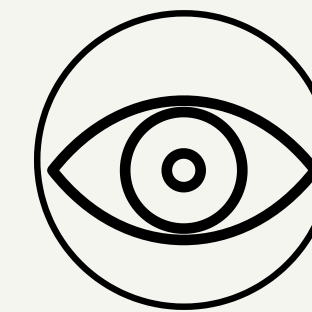


Visuo-proprioceptive Conflict

Human brain combines sensory signals by weighting them according to their reliability.

In spatial interactions, vision often has a strong influence over proprioception. By exploiting this visual dominance, controlled **visuo-proprioceptive conflicts** can be created to alter the user's perceived haptic properties.

By **altering the Control/Display ratio** between the actual physical semi-squeeze movement - based on palm and thumb interaction - and the extent of the visual deformation of the virtual ball across three gain levels (low, medium, and high), the study aims to evaluate whether visual deformation alone can generate a convincing pseudo-haptic illusion of softness.



VISION

HAND POSITION



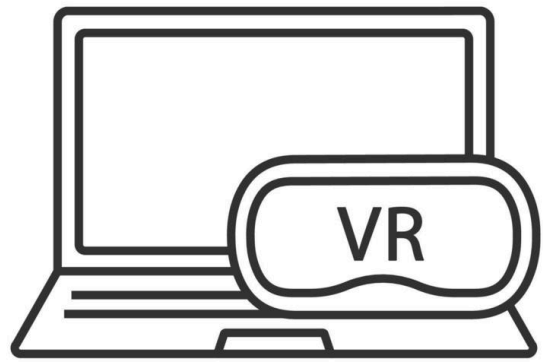
PROPRIOCEPTION

PERCEIVED POSITION

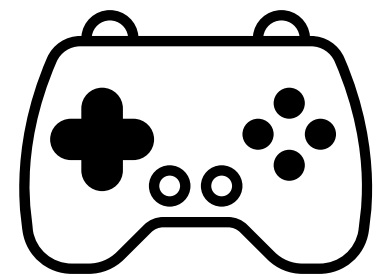


CONFLICT

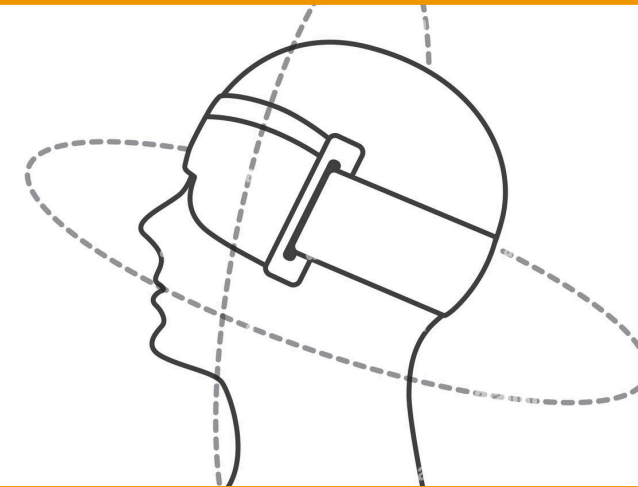
State of the Art



ENVIRONMENT



CURRENT
LIMITS



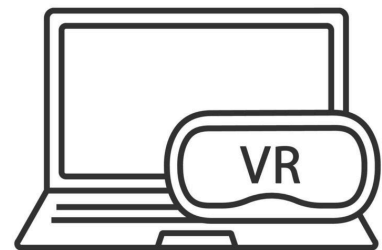
RESEARCH
GAP



PROJECT

State of the Art

ENVIRONMENT



Most of the research on pseudo-haptic feedback has been conducted within immersive Virtual Reality (VR).

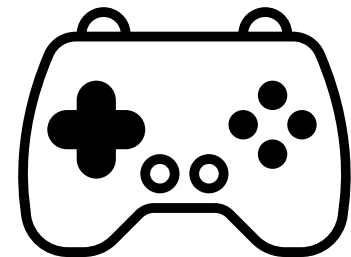
CURRENT
LIMITS

RESEARCH
GAP

PROJECT

State of the Art

CURRENT LIMITS



Many existing studies are based on desktop interfaces, the use of props that the user must hold, or take place in highly constrained workspaces.

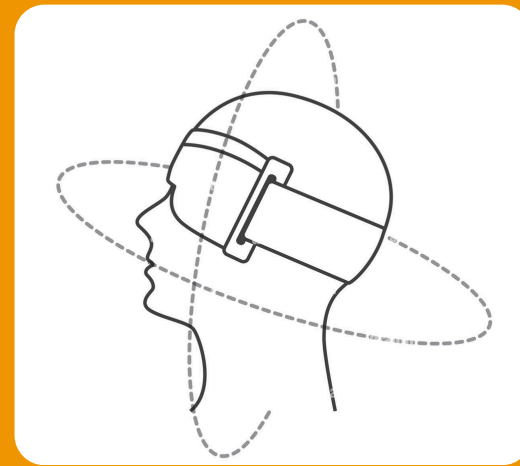
ENVIRONMENT

RESEARCH
GAP

PROJECT

State of the Art

RESEARCH GAP



There is a lack of systematic exploration of pseudo-optic illusions in an Optical See-Through Augmented Reality (OST-AR) environment, using natural mid-air hand interaction.

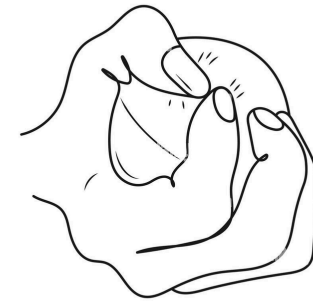
ENVIRONMENT

CURRENT
LIMITS

PROJECT

State of the Art

PROJECT



Use AR (Magic Leap 2) and natural finger tracking, basing the entire illusion on the visuo-proprioceptive conflict without any additional device, focused on the sensation of softness.

ENVIRONMENT

CURRENT
LIMITS

RESEARCH
GAP



Research Question

Can visual manipulation alone induce a convincing pseudo-haptic perception of softness during bare-handed interaction in mid-air AR?

Methodology

POPULATION

13 participants aged between 22-28 years.
Most are right-handed with little or no previous experience with AR.

INTERVENTION

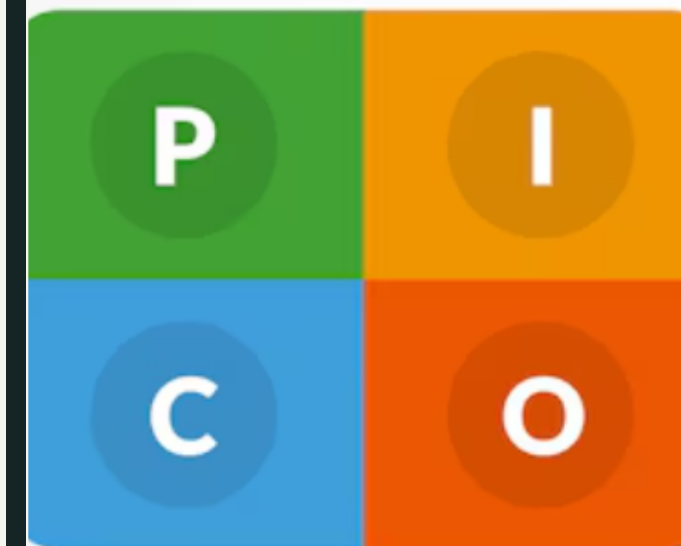
Manipulation of visual **gain** of deformation.

- low (0.6)
- medium (1.0)
- high (1.4)

COMPARISON

Between the gain level:

- low
- medium
- high



OUTCOME

Subjective perception and objective interaction metrics.

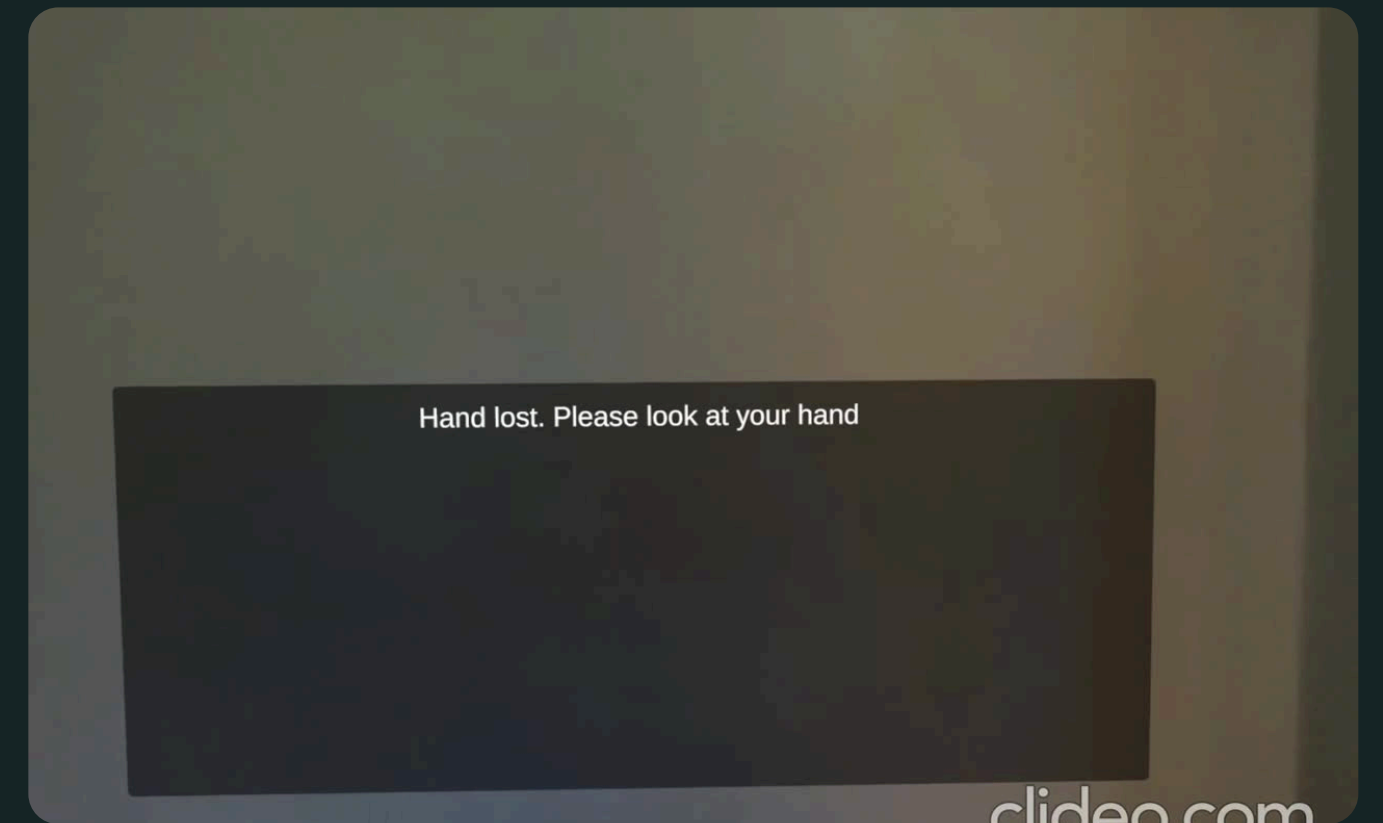
Experimental Design

The experiment consisted of **six trials**, two for each level of deformation gain: low, medium, and high. The trials were preceded by a **calibration phase** and followed by a **post trials questionnaire**.



CALIBRATION

- collect data to describe the sample
- collected useful hand tracking data, such as the distance between the thumb and the palm and the squeeze movement



[Click the link to view the video](#)

TRIALS

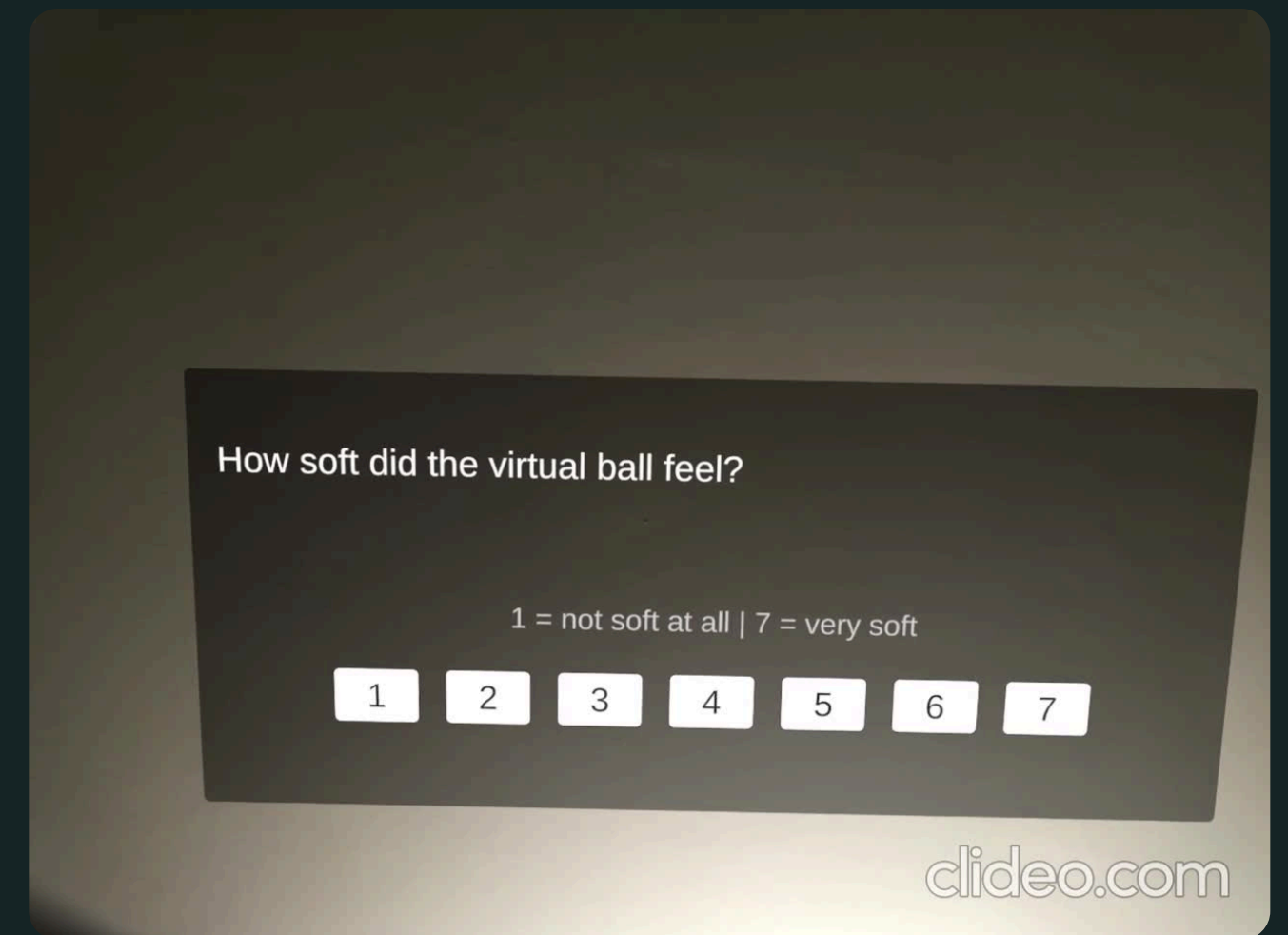
- Each participant had 13 seconds for each trial to interact with the virtual ball
- At the end of each trial, the participant was asked to answer the question: "How clearly did you perceive the deformation of the ball?". The answer was given using a 7-point Likert scale.



[Click the link to view the video](#)

POST TRIALS QUESTIONNAIRE

- Questions Q1 to Q7 used a 7-point Likert scale.
- Question Q8 was introduced to verify the threshold for the limit of the illusion.



[Click the link to view the video](#)

SENSOR DEVICE:

Magic Leap tracking acts as a data producer, encapsulating the data in the hardware and exposing it to the Unity system.

Cognitive Approach

DESIGN PATTERNS

ASYNCHRONOUS LOGIC:

The module that saves the data to CSV at the end of the trial subscribes to the data without blocking the visual rendering.

COMPUTATIONAL:

The Unity scripts represent the module that calculates the deformation. Unity receives finger data as input, updates the pseudo-haptic logic, and publishes the visual transformation as output.

SENSE:

Magic Leap reads hand tracking data and contact (collisions) with the ball.

Cognitive Approach

HIERARCHICAL ARCHITECTURES

PLAN/REASONING:

C# scripts evaluates the current gain level (Low/Med/High) and calculates the mathematical deformation based on the hand-movement.

ACT:

Unity applies the blend shape and visual deformation on the virtual ball.

LEVEL 1 - SUBSUMPTION:

The pseudo-optic behavior is activated, inhibiting natural tracking and applying visual gain by altering the C/D ratio until contact ends.

Cognitive Approach

BIO-INSPIRED ARCHITECTURES

RELEASER:

User comes into contact with the ball.

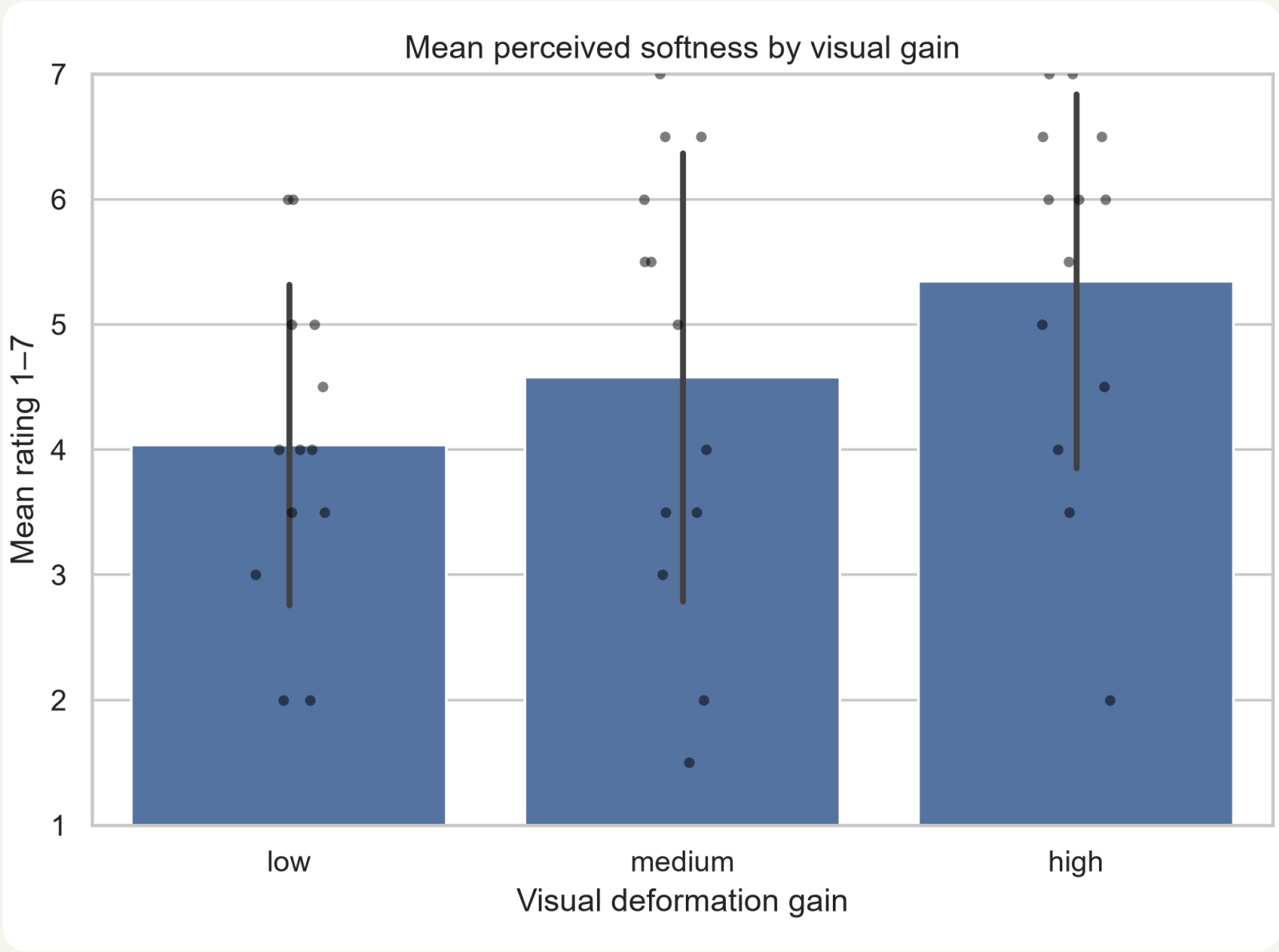
LEVEL 0 - REFLEX:

Free movement. The virtual hand traces the real hand.

Results

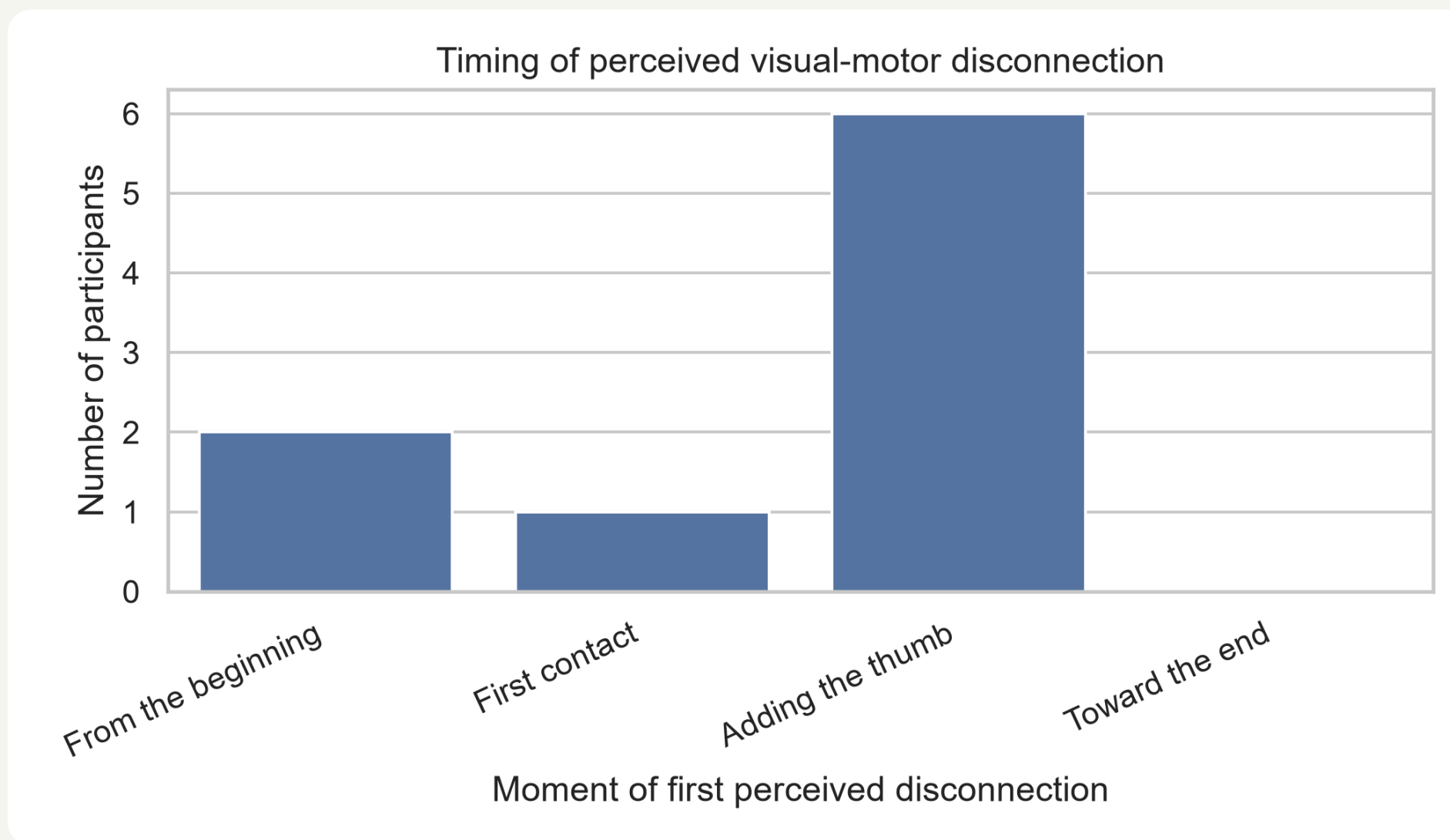
By visualizing the mean ratings of the trial-data, it is possible to observe that the **high deformation gain produced the clearest perceived deformation** according to participant ratings.

The Friedman statistical test ($\chi^2 = 11.91$ and $p\text{-value}=0.003$) indicates a significant difference between gain levels. Post-hoc tests (Wilcoxon and Conover) confirmed that the **high-gain** condition showed the strongest perceptual difference, while low vs. medium was not significant.



Factor	X ² _F	df	p	Kendall's W
Deformation L _E vel	11.91	2	.003	0.4581

Visual Disconnection



9 out of 13 participants say they perceived a visual disconnection during the experiment, 69.2% report having perceived it with the addition of the thumb.

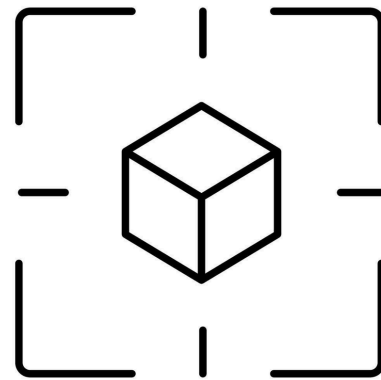
We can therefore identify the **thumb as a limitation** of the current prototype.

Future Developments



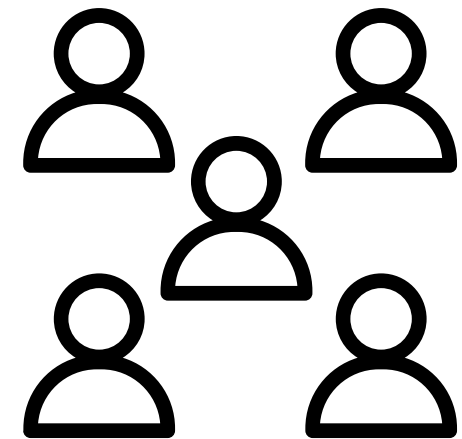
THUMB CONTACT

Improvement of illusion deformation during the thumb contact



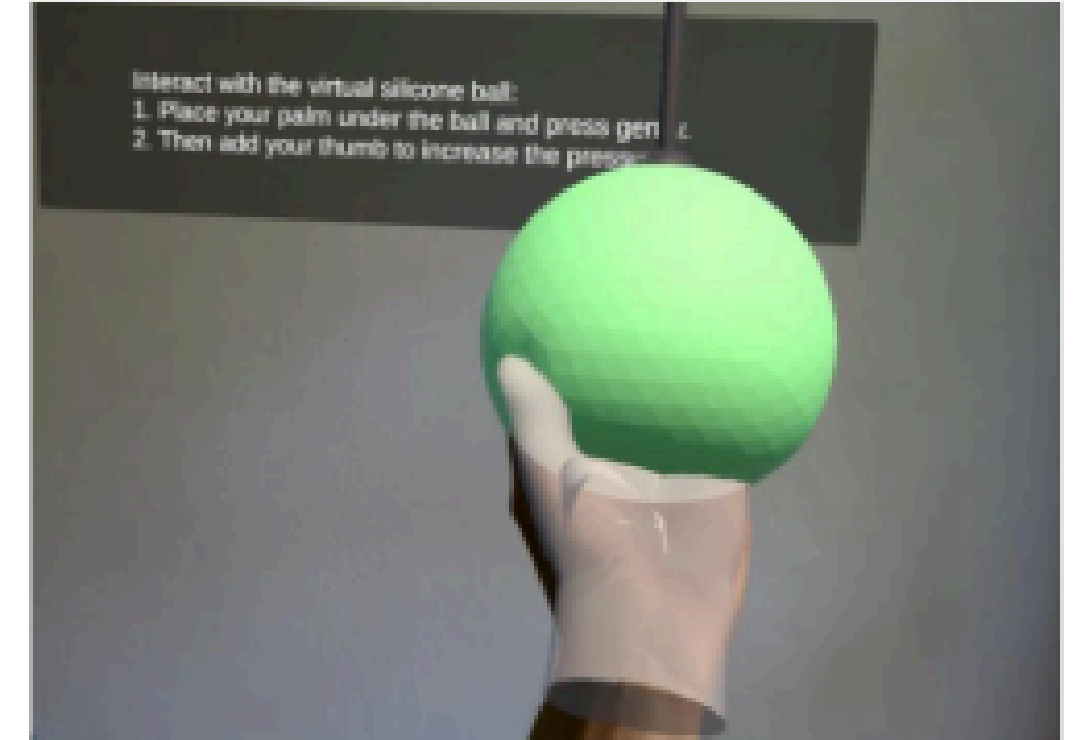
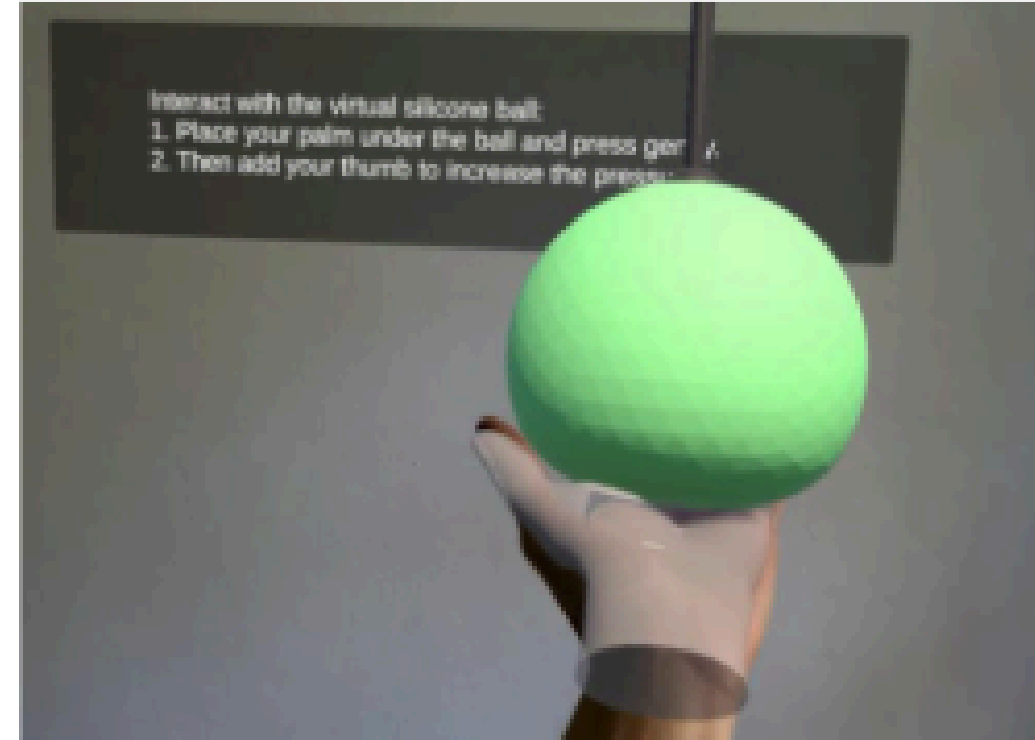
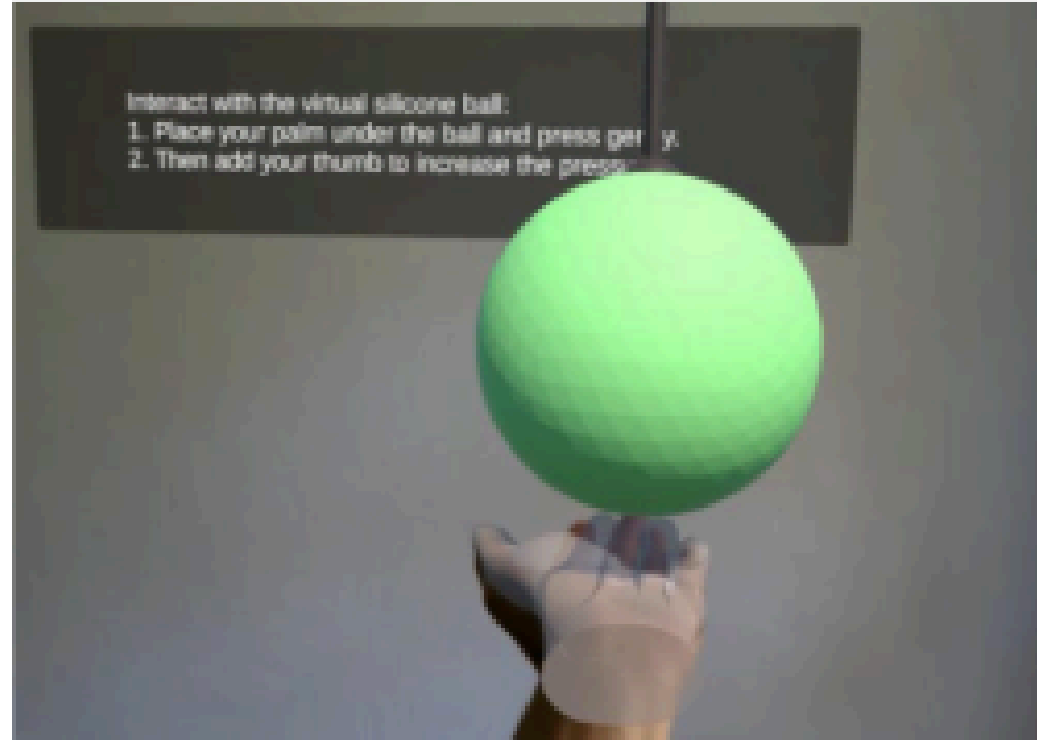
OPTICAL TRACKING

Reduction of the temporary noise in the tracking data and improve the fluidity of the mesh response



SAMPLE AND TEST

Broader and more diverse sample.
Adding psychophysical methods (JND/PSE)



Thank You

QUESTIONS?

- s8434670@studenti.unige.it
- https://github.com/Chiaera/Cognitive_Architectures.git